Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.02	20.13	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.01	+0.6%
power (W), mean	627	639	+1.8%
power (W), peak	657	676	+3.0%
energy (Wh)	209	214	+2.4%
energy, parked workers still on at idle power (Wh)	209	214	+2.4%
CPU used (cores), mean	0.841	1.14	+35.5%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.035	0.047	+35.5%
energy per core-hour (Wh)	746	561	-24.8%
node-hours per core-hour	7.14	5.27	-26.2%
pending pods, pod-minutes	0.000	0.267	new
pending pods, peak	0.000	1.00	new
HPA replicas, mean	9.04	9.30	+2.9%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.517	-34.0%
requests timed	1,055	1,115	+5.7%
time over the response line (% of samples)	1.04	0.090	-91.4%
response time (ms), mean	139	80.62	-41.8%
response time (ms), median	115	73.10	-36.5%
response time (ms), 95th percentile	277	111	-60.0%
response time (ms), 99th percentile	499	149	-70.2%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.8457	6.0251	-23.2%	[-3.6855, +0.0484]	EVEN, no significant difference
kwh_per_core_hour	0.8164	0.6358	-22.1%	[-0.3549, +0.0014]	EVEN, no significant difference
utilisation	0.0346	0.0467	+35.1%	[+0.0005, +0.0247]	GAIN
pending_pod_minutes_per_hour	0.0000	0.7619	n/a	[+0.0000, +2.2857]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.3041	1.4401	-37.5%	[-3.8403, +2.0876]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.359 0.392 0.421 0.446 0.469 0.488 0.505 0.521 0.534 0.546 0.556 0.561 0.566 0.571 0.576 0.58 0.584 0.587 0.589 0.59
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	619	5.50	6.00	0.000	0.000	0.546	0.582

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	617	622	6.00	6.50	0.000	0.000	0.549	0.695
2	6.00	6.00	616	623	6.00	7.00	0.000	0.000	0.538	0.725
3	6.00	6.00	620	628	6.00	7.00	0.000	0.000	0.648	0.876
4	6.00	6.00	632	650	8.00	10.00	0.000	0.000	0.953	1.36
5	6.00	6.00	633	646	10.00	10.00	0.000	0.000	1.00	1.32
6	6.00	6.00	630	643	10.00	10.00	0.000	0.000	0.919	1.27
7	6.00	6.00	644	670	10.00	10.00	0.000	0.000	1.28	1.99
8	6.00	6.00	647	664	10.00	10.00	0.000	0.000	1.34	1.80
9	6.00	6.00	642	668	10.00	10.00	0.000	0.000	1.26	1.93
10	6.00	6.00	635	665	10.00	10.00	0.000	0.000	1.08	1.84
11	6.00	6.00	617	624	10.00	10.00	0.000	0.000	0.570	0.753
12	6.00	6.00	616	620	10.00	10.00	0.000	0.000	0.598	0.670
13	6.00	6.00	621	631	10.00	10.00	0.000	0.250	0.679	0.930
14	6.00	6.00	633	638	10.00	10.00	0.000	0.000	0.974	1.14
15	6.00	6.00	630	644	10.00	10.00	0.000	0.000	0.949	1.32
16	6.00	6.00	632	645	10.00	10.00	0.000	0.000	0.963	1.29
17	6.00	6.00	621	636	10.00	10.00	0.000	0.000	0.708	1.06
18	6.00	6.00	616	625	10.00	10.00	0.000	0.000	0.543	0.772
19	6.00	6.00	618	620	10.00	10.00	0.000	0.000	0.577	0.678

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.